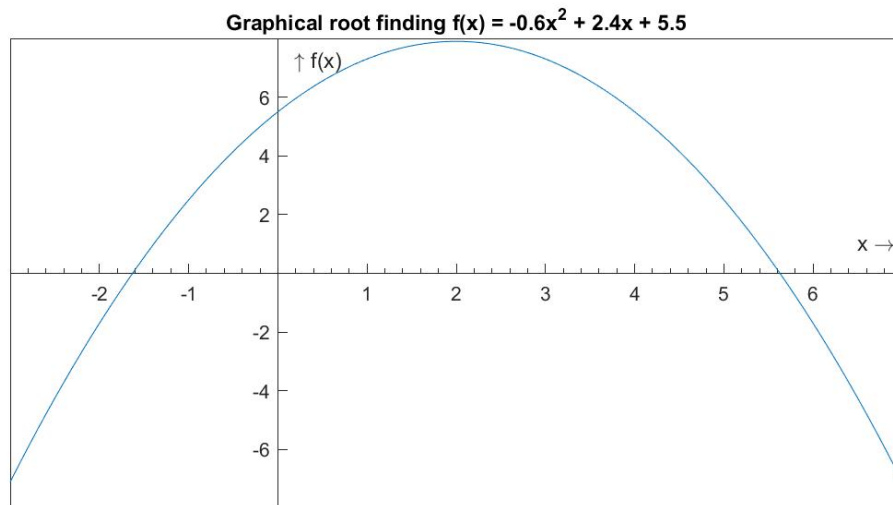


MTE 204 Problem Set #1

Roots of Equations (Chapter 5, 6, 8 in Chapra 8th edition, International Version)

5.1 Determine the real roots of $f(x) = -0.6x^2 + 2.4x + 5.5$:

- (a) Graphically.
 - (b) Using the quadratic formula.
 - (c) Using three iterations of the bisection method to determine the highest root. Employ initial guesses of $x_l = 5$ and $x_u = 10$. Compute the estimated error ε_a and the true error ε_t after each iteration.
- (a) A plot indicates that roots occur at about $x = -1.6$ and 5.6 .



(b)

$$\varepsilon_a = \left[\frac{x_r^{\text{new}} - x_r^{\text{old}}}{x_r^{\text{new}}} \right] \cdot 100\%$$

Relative error Approximate

$$\varepsilon_a = (7.5 - 5) / 7.5 \times 100\% = 33.33\%$$

$$x = \frac{-2.4 \pm \sqrt{(2.4)^2 - 4(-0.6)(5.5)}}{2(-0.6)} = \begin{matrix} -1.62859 \\ 5.62859 \end{matrix}$$

(c) First iteration:

$$x_r = \frac{5 + 10}{2} = 7.5$$

$$\varepsilon_t = \left| \frac{5.62859 - 7.5}{5.62859} \right| \times 100\% = 33.24\% \quad \varepsilon_a = \left| \frac{10 - 5}{10 + 5} \right| \times 100\% = 33.33\%$$

This is eqn 5.3 in Chapra 8th ed.

$$f(5)f(7.5) = 2.5(-10.5) = -26.25$$

Therefore, the bracket is $x_l = 5$ and $x_u = 7.5$.

Second iteration:

$$x_r = \frac{5 + 7.5}{2} = 6.25$$

$$\varepsilon_t = \left| \frac{5.62859 - 6.25}{5.62859} \right| \times 100\% = 11.04\% \quad \varepsilon_a = \left| \frac{7.5 - 5}{7.5 + 5} \right| \times 100\% = 20.00\%$$

$$f(5)f(6.25) = 2.5(-2.9375) = -7.3438$$

Consequently, the new bracket is $x_l = 5$ and $x_u = 6.25$.

Third iteration:

$$x_r = \frac{5 + 6.25}{2} = 5.625$$

$$\varepsilon_t = \left| \frac{5.62859 - 5.625}{5.62859} \right| \times 100\% = 0.0638\% \quad \varepsilon_a = \left| \frac{6.25 - 5}{6.25 + 5} \right| \times 100\% = 11.11\%$$

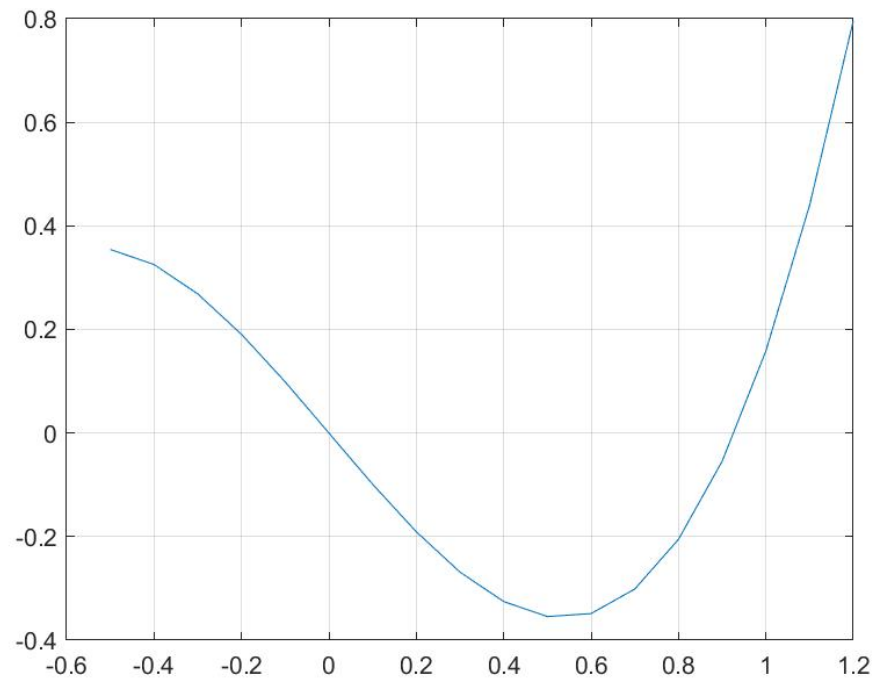
Or, in tabular form (note that we use capital L and U in Excel and Matlab because the lower case l and number 1 appear very similar in several common fonts):

xL	xr	xU	fxL	fxr	fxU	fxL*fxr	ea(%)	et(%)
5	7.5	10	2.5	-10.25	-30.5	-25.625	33.33333	33.24829
5	6.25	7.5	2.5	-2.9375	-10.25	-7.34375	20	11.04024
5	5.625	6.25	2.5	0.015625	-2.9375	0.039063	11.11111	0.063785

5.5 Locate the first nontrivial root of $\sin x = x^3$ where x is in radians. Use a graphical technique and bisection with the initial interval from 0.5 to 1. Perform the computation until ε_a is less than $\varepsilon_s = 2\%$. Also perform an error check by substituting your final answer into the original equation.

A graph of the function can be generated with MATLAB interactively

```
>> x=[-0.5:0.1:1.2];
>> f= x.^3 - sin(x);
>> plot(x,f)
>> grid
```



Scripting can be used to change default plotting parameters and add chart elements, though the few lines above are sufficient to solve the problem:

```
% Chapra 8e Chapter 5 problem 5
% RJ Genik II, 15 December 2019

clear

hold off

f = @(x) x.^3 - sin(x);
xtrue = fzero(f,0.6)

xmin = -0.5; xmax = 1.2;

% part a)

x = linspace(xmin, xmax);
y = f(x);

set(gcf,'Position',[400 400 800 400])

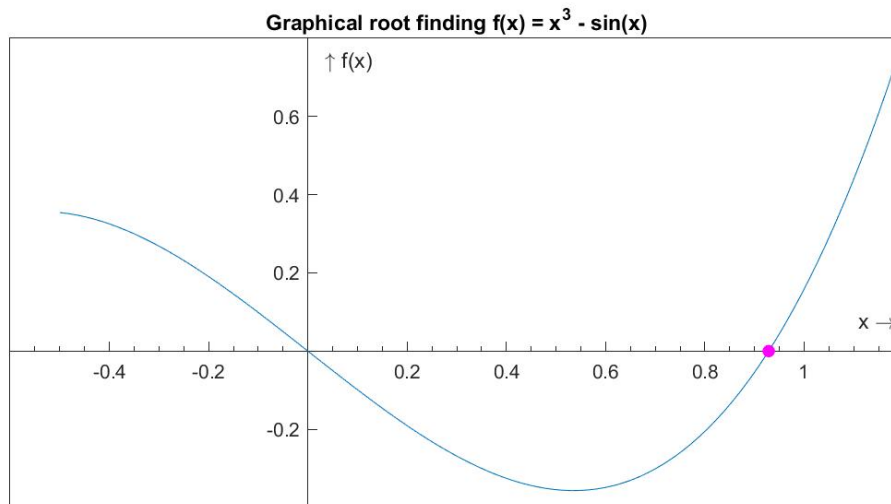
plot(x,y)

hold on
```

```

plot(xtrue,0,'mo','MarkerFaceColor','m')
set(gca, 'XAxisLocation','origin','YAxisLocation','origin','XMinorTick','on')
xlabel('x \rightarrow')
ylabel('\uparrow f(x)')
title('Graphical root finding f(x) = x^3 - sin(x)')
saveas(gcf,'ch5p5a.jpg')

```



Note that both functions

$$f(x) = x^3 - \sin(x) \quad f(x) = \sin(x) - x^3$$

have the same root(s). These plots indicate that a nontrivial root (i.e., nonzero) is located at about 0.925.

Using bisection, the first iteration is

$$x_r = \frac{0.5+1}{2} = 0.75$$

$$f(0.5)f(0.75) = -0.35443(-0.25976) = 0.092067$$

Therefore, the root is in the second interval and the lower guess is redefined as $x_l = 0.75$. The second iteration is

$$x_r = \frac{0.75+1}{2} = 0.875$$

$$\varepsilon_a = \left| \frac{0.875 - 0.75}{0.875} \right| 100\% = 14.29\%$$

$$f(0.75)f(0.875) = -0.25976(-0.09762) = 0.025359$$

Because the product is positive, the root is in the second interval and the lower guess is redefined as $x_l = 0.875$. The remainder of the iterations are displayed in the following table:

i	x_l	x_r	x_u	$f(x_u)$	$f(x_r)$	$f(x_l)$	$ \varepsilon_a $ (%)
1	0.5	0.75	1	-0.35443	-0.25976	0.158529	33.33
2	0.75	0.875	1	-0.25976	-0.09762	0.158529	14.29
3	0.875	0.9375	1	-0.09762	0.017894	0.158529	6.67
4	0.875	0.90625	0.9375	-0.09762	-0.0429	0.017894	3.45
5	0.90625	0.921875	0.9375	-0.0429	-0.01328	0.017894	1.69

Therefore, after five iterations we obtain a root estimate of **0.921875** with an approximate error of 1.7%, which is below the stopping criterion of 2%. As in the above table, the function value at the root estimate is -0.013277424 .

5.14 Use bisection to determine the drag coefficient needed so that an 80-kg parachutist has a velocity of 35 m/s after 3.9 s of free fall. Note: The acceleration due to gravity is 9.81 m/s^2 . Start with initial guesses of $x_l = 3$ and $x_u = 5$ and iterate until the approximate relative error falls below 5%. Also perform an error check by substituting your final answer into the original equation.

The function to evaluate is

$$f(c) = \frac{9.81(82)}{c} \left(1 - e^{-(c/82)^4} \right) - 36 = 0$$

$$f(c) = 9.81 * 80/c * (1 - \exp(-c/80 * 3.9)) - 35$$

[There is a slight misprint in the solutions manual. In the above equation, which is a rearrangement of the example we did in class, the mass should be 80 (not 82) and the velocity should be 35 m/s (not 36 m/), and time is 3.9s. further down in the solution (below the table), the correct values are shown. You are solving for 'c' in this example.]

The first iteration is

$$x_r = \frac{3+5}{2} = 4$$

$$f(3)f(4) = 0.592854(-0.24016) = -0.142380$$

Therefore, the root is in the first interval and the upper guess is redefined as $x_u = 4$. The second iteration is

$$x_r = \frac{3+4}{2} = 3.5$$

$$\varepsilon_a = \left| \frac{3.5-4}{3.5} \right| 100\% = 14.29\%$$

$$f(3)f(3.5) = 0.592854(0.173011) = 0.102571$$

Therefore, the root is in the upper interval and the lower bound is redefined as $x_l = 3.5$. The remaining iterations are displayed in the following table:

i	x_l	x_r	x_u	$f(x_l)$	$f(x_r)$	$f(x_u)$	$f(x_l) \times f(x_r)$	$ \varepsilon_a (\%)$
1	3	4	5	0.592854	-0.24016	-1.04696697	-0.142380	25.00
2	3	3.5	4	0.592854	0.173011	-0.24015991	0.102571	14.29
3	3.5	3.75	4	0.173011	-0.034401	-0.24015991	-0.005952	6.67
4	3.5	3.625	3.75	0.173011	0.069098	-0.03440052	0.011955	3.45
5	3.625	3.6875	3.75	0.069098	0.017297	-0.03440052	0.001195	1.69

Thus, after five iterations, we obtain a root estimate of **3.6875** with an approximate error of 1.69%. This result can be checked by substituting your final answer into the original equation to yield

$$v = \frac{9.81(80)}{3.6875} \left(1 - e^{-(3.6875/80)3.9} \right) = 35.0173$$

Note that this result is within 0.05% of the input velocity.

6.2 Determine the highest real root of

$$f(x) = 2.1x^3 - 11.6x^2 + 17.5x - 6$$

(a) Graphically.

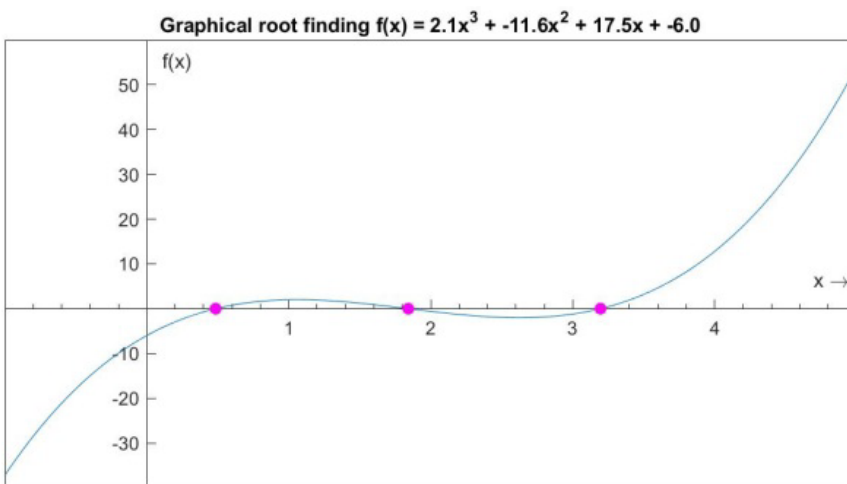
(b) Fixed-point iteration method (three iterations, $x_0 = 3$). Note: Make certain that you develop a solution that converges on the root.

(c) Newton-Raphson method (three iterations, $x_0 = 3$).

(d) Secant method (three iterations, $x_{-1} = 3, x_0 = 4$).

Compute the approximate percent relative errors for your solutions.

(a) Graphical



Root ≈ 3.2

(b) Fixed point

The equation can be solved in numerous ways. A simple way that converges is to solve for the x that is not raised to a power to yield

$$x = \frac{6 - 2.1x^3 + 11.5x^2}{17.5}$$

The resulting iterations are

i	x_i	ε_a (%)
0	3	
1	3.068571	2.234637
2	3.117118	1.557426
3	3.14899	1.012117

(c) Newton-Raphson

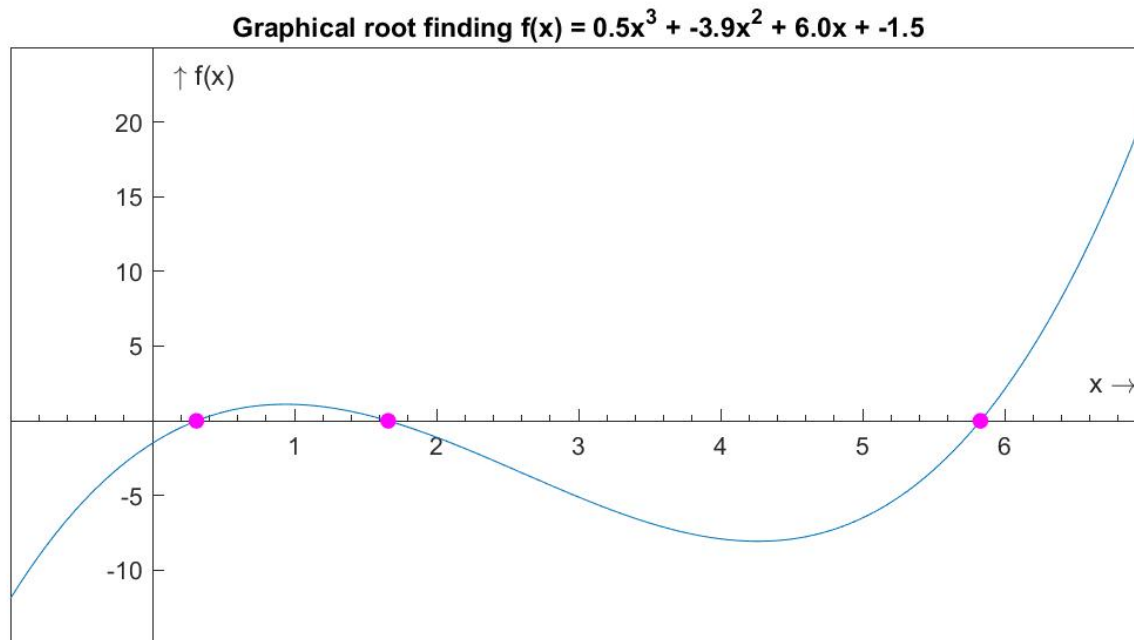
i	x_i	$f(x)$	$f'(x)$	ε_a (%)
0	3	-1.2	4.6	
1	3.26087	0.534068	8.837429	8
2	3.200437	0.032199	7.779484	1.888256
3	3.196298	0.000147	7.708711	0.129492

(d) Secant

i	x_i	$f(x_i)$	ε_a (%)
-1	3	-1.2	
0	4	12.8	
1	3.085714	-0.75076	25
2	3.136369	-0.43162	29.62963
3	3.204878	0.066916	1.615072

6.4 Determine the real roots of $f(x) = -1.5 + 6x - 3.9x^2 + 0.5x^3$: (a) graphically and (b) using the Newton-Raphson method to within $\varepsilon_s = 0.01\%$.

(a) A graph of the function indicates that there are 3 real roots at approximately 0.3, 1.7, and 5.8.



(b) The Newton-Raphson method can be set up as

$$x_{i+1} = x_i - \frac{-1.5 + 6x_i - 3.9x_i^2 + 0.5x_i^3}{6 - 7.8x_i + 1.5x_i^2}$$

This formula can be solved iteratively to determine the three roots as summarized in the following tables:

i	x_i	$f(x)$	$f'(x)$	$\epsilon_a(\%)$
0	0	-1.5	6	
1	0.25	-0.23594	4.14375	100
2	0.306938	-0.01134	3.747199	18.55037
3	0.309963	-3.1E-05	3.726403	0.975949
4	0.309972	-2.4E-10	3.726345	0.002724

i	x_i	$f(x)$	$f'(x)$	$\epsilon_a(\%)$
0	2	-1.1	-3.6	

1	1.694444	-0.09829	-2.90995	18.03279
2	1.660667	-0.00157	-2.81648	2.033989
3	1.66011	-4.4E-07	-2.81491	0.033558
4	1.660109	-3.6E-14	-2.81491	9.36E-06

i	x_i	$f(x)$	$f'(x)$	$\epsilon_a(\%)$
0	6	2.1	13.2	
1	5.840909	0.127067	11.61524	2.723735
2	5.829969	0.000581	11.50905	0.187646
3	5.829919	1.24E-08	11.50856	0.000866

Therefore, the roots are 0.309972, 1.660109, and 5.829919.

6.11 Use the Newton-Raphson method to find the root of

$$f(x) = e^{-0.5x}(4-x) - 2$$

Employ initial guesses of **(a)** 2, **(b)** 6, and **(c)** 8. Explain your results.

The Newton-Raphson method can be set up as

$$x_{i+1} = x_i - \frac{e^{-0.5x_i}(4-x_i) - 2}{-e^{-0.5x_i}(3-0.5x_i)}$$

(a)

i	x	$f(x)$	$f'(x)$
0	2	-1.26424	-0.73576
1	0.281718	1.229743	-2.48348
2	0.776887	0.18563	-1.77093

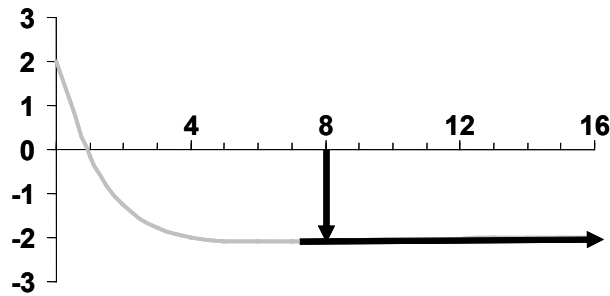
3	0.881708	0.006579	-1.64678
4	0.885703	9.13E-06	-1.64221
5	0.885709	1.77E-11	-1.6422
6	0.885709	0	-1.6422

(b) The case does not work because the derivative is zero at $x_0 = 6$.

(c)

i	x	$f(x)$	$f'(x)$
0	8	-2.07326	0.018316
1	121.1963	-2	2.77E-25
2	7.21E+24	-2	0

This guess breaks down because, as depicted in the following plot, the near zero, positive slope sends the method away from the root.



6.19 You are designing a spherical tank (Fig. P6.19) to hold water for a small village in a developing country. The volume of liquid it can hold can be computed as

$$V = \pi h^2 \frac{[3R - h]}{3}$$

where V = volume (m^3), h = depth of water in tank (m), and R = the tank radius (m). If $R = 4$ m, what depth must the tank be filled to so that it holds 50 m^3 ? Use three iterations of the Newton-Raphson method to determine your answer. Determine the approximate relative error after each iteration. Note that an initial guess of R will always converge.

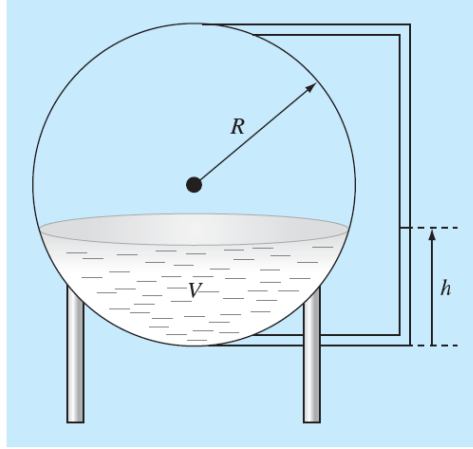


FIGURE P6.19

The equation to be solved is

$$f(h) = \pi R h^2 - \left(\frac{\pi}{3}\right) h^3 - V$$

Because this equation is easy to differentiate, the Newton-Raphson is the best choice to achieve results efficiently. It can be formulated as

$$x_{i+1} = x_i - \frac{\pi R x_i^2 - \left(\frac{\pi}{3}\right) x_i^3 - V}{2\pi R x_i - \pi x_i^2}$$

or substituting the parameter values,

$$x_{i+1} = x_i - \frac{\pi(4)x_i^2 - \left(\frac{\pi}{3}\right)x_i^3 - 50}{2\pi(4)x_i - \pi x_i^2}$$

The iterations can be summarized as

iteration	x_i	$f(x_i)$	$f'(x_i)$	$ \mathcal{E}_a $ (%)
0	4	84.04129	50.26548	
1	2.328052	4.894374	41.48344	71.81749
2	2.210068	0.074837	40.20027	5.338469
3	2.208206	1.95E-05	40.17932	0.084304

Thus, after only three iterations, the root is determined to be 2.208206 with an approximate relative error of 0.084%.

8.4 The following equation pertains to the concentration of a chemical in a completely mixed reactor:

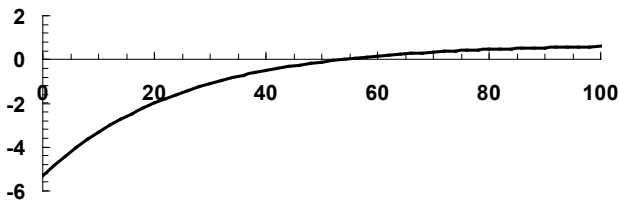
$$c = c_{\text{in}} (1 - e^{-0.04t}) + c_0 e^{-0.04t}$$

If the initial concentration $c_0 = 4$ and the inflow concentration $c_{\text{in}} = 10$, compute the time required for c to be 93% of c_{in} .

The function to be solved is

$$f(t) = 10(1 - e^{-0.04t}) + 4e^{-0.04t} - 9.3 = 0$$

A plot of the function indicates a root at about $t = 55$.



Bisection with initial guesses of 0 and 60 can be used to determine a root of 53.711 after 16 iterations with $\varepsilon_a = 0.002\%$.

8.44 The upward velocity of a rocket can be computed with the following formula:

$$v = u \ln \frac{m_0}{m_0 - qt} - gt$$

where v = upward velocity, u = the velocity at which fuel is expelled relative to the rocket, m_0 = the initial mass of the rocket at time $t = 0$, q = the fuel consumption rate, and g = the downward acceleration of gravity (assumed constant = 9.81 m/s^2). If $u = 2200 \text{ m/s}$, $m_0 = 160,000 \text{ kg}$, and $q = 2680 \text{ kg/s}$, compute the time at which $v = 1000 \text{ m/s}$. (Hint: t is somewhere between 10 and 50 s.) Determine your result so that it is within 1% of the true value. Check your answer.

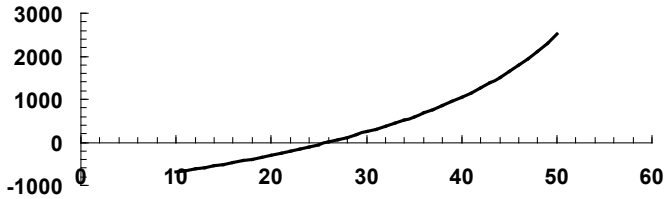
The solution can be formulated as

$$f(t) = u \ln \frac{m_0}{m_0 - qt} - gt - v$$

Substituting the parameter values gives

$$f(t) = 2,200 \ln \frac{160,000}{160,000 - 2,680t} - 9.81t - 1,000$$

A plot of this function indicates a root at about $t = 26$.



Because two initial guesses are given, a bracketing method like bisection can be used to determine the root,

i	t_l	t_u	t_r	$f(t_l)$	$f(t_r)$	$f(t_l) \times f(t_r)$	ϵ_a
1	10	50	30	-694.791	241.6514	-167897	
2	10	30	20	-694.791	-298.67	207513.3	50.00%
3	20	30	25	-298.67	-51.5865	15407.33	20.00%
4	25	30	27.5	-51.5865	88.38228	-4559.33	9.09%
5	25	27.5	26.25	-51.5865	16.86085	-869.792	4.76%
6	25	26.25	25.625	-51.5865	-17.7329	914.7789	2.44%
7	25.625	26.25	25.9375	-17.7329	-0.53026	9.403139	1.20%
8	25.9375	26.25	26.09375	-0.53026	8.141517	-4.31716	0.60%

Thus, after 8 iterations, the approximate error falls below 1% with a result of $t = 26.09375$. Note that if the computation is continued, the root can be determined as 25.94708.

